



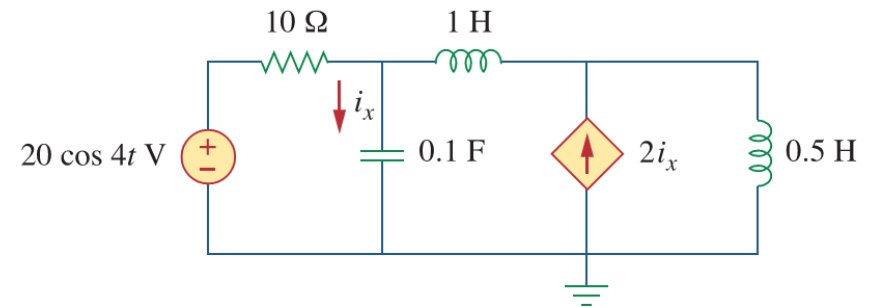
Lecture 9

- Sinusoidal Steady-State Analysis



Outline

- Kirchhoff's laws in phasor domain
- Generalized impedance
- General AC phasor analysis
- Phasor diagram





Kirchhoff's Laws in the Phasor Domain

- Let $v_1, v_2, \dots, v_n$ be the voltages around a closed loop.
Then according to KVL

$$v_1 + v_2 + \dots + v_n = 0$$

$$\mathbf{V}_1 + \mathbf{V}_2 + \dots + \mathbf{V}_n = 0$$

Similarly, KCL holds for phasors:

$$i_1 + i_2 + \dots + i_n = 0$$

$$\mathbf{I}_1 + \mathbf{I}_2 + \dots + \mathbf{I}_n = 0,$$

Proof

If

$$v_1 + v_2 + \cdots + v_n = 0$$

where v_i are sinusoidal voltages of the same frequency, then

$$\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n = 0$$

$$v_1 + v_2 + \cdots + v_n = 0$$



$$V_{m1} \cos(\omega t + \theta_1) + V_{m2} \cos(\omega t + \theta_2) + \cdots + V_{mn} \cos(\omega t + \theta_n) = 0$$



$$\operatorname{Re}(V_{m1} e^{j\theta_1} \cdot e^{j\omega t}) + \cdots + \operatorname{Re}(V_{mn} e^{j\theta_n} \cdot e^{j\omega t}) = 0$$



$$\operatorname{Re}((\mathbf{V}_1 + \cdots + \mathbf{V}_n) \cdot e^{j\omega t}) = 0 \quad \text{Where } \mathbf{V}_k = V_{mk} e^{j\theta_k}$$



$$\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n = 0$$



Outline

- Kirchhoff's laws in phasor domain
- Generalized impedance
- General AC phasor analysis
- Phasor diagram



Review: Impedance and Admittance

Resistor	$\mathbf{Z} = R$	$\mathbf{Y} = 1/R$
Inductor	$\mathbf{Z} = j\omega L$	$\mathbf{Y} = 1/j\omega L$
Capacitor	$\mathbf{Z} = 1/j\omega C$	$\mathbf{Y} = j\omega C$

Impedance is
voltage/current

$$\mathbf{Z} = R + jX$$

R = resistance = $\text{Re}(\mathbf{Z})$

X = reactance = $\text{Im}(\mathbf{Z})$

Admittance is
current/voltage

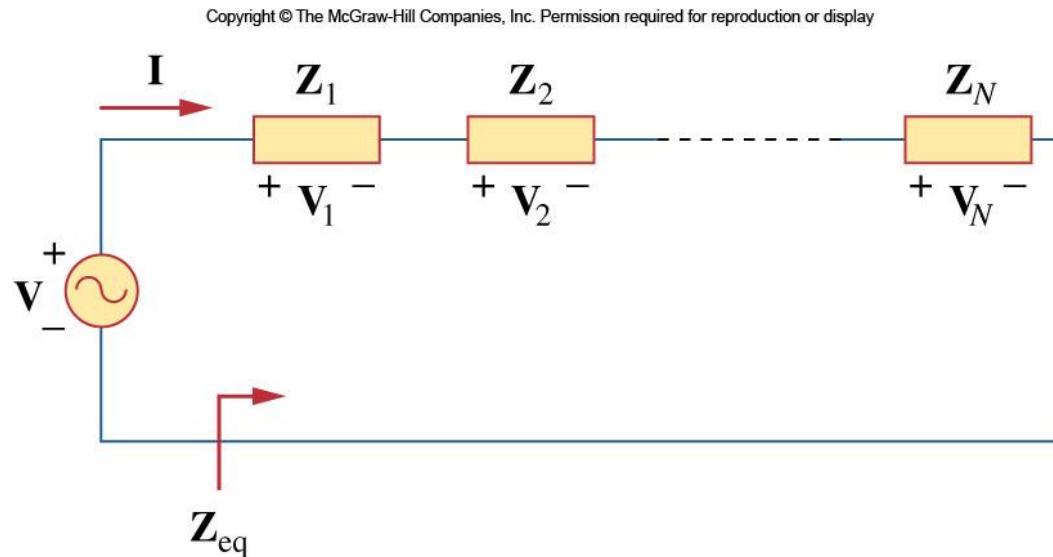
$$\mathbf{Y} = \frac{1}{\mathbf{Z}} = G + jB$$

G = conductance = $\text{Re}(\mathbf{Y})$

B = susceptance = $\text{Im}(\mathbf{Y})$

Series Impedance

- Once in frequency domain, the impedance elements are generalized, combinations will follow the rules for resistors:

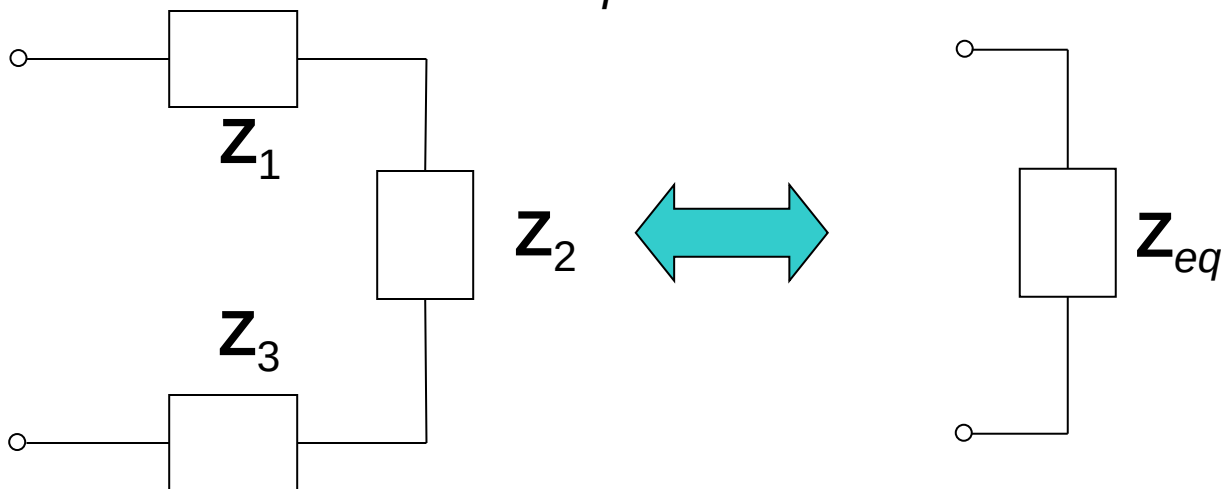


$$Z_{eq} = Z_1 + Z_2 + Z_3 + \cdots + Z_N$$

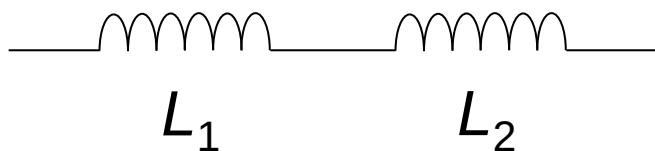


Series Impedance

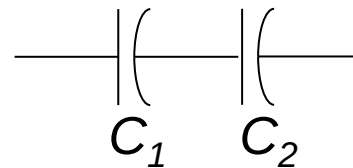
$$\mathbf{Z}_{eq} = \mathbf{Z}_1 + \mathbf{Z}_2 + \mathbf{Z}_3$$



For example:



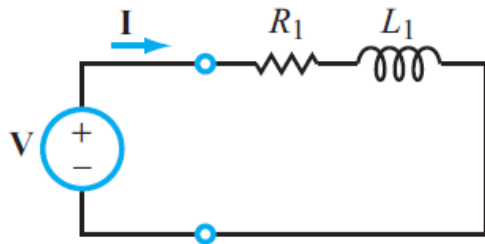
$$\mathbf{Z}_{eq} =$$



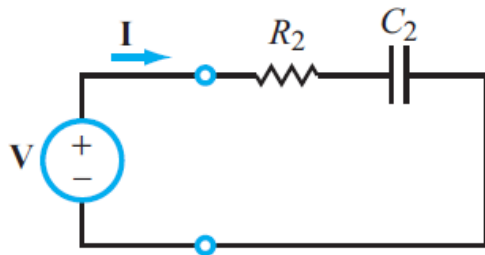
$$\mathbf{Z}_{eq} =$$



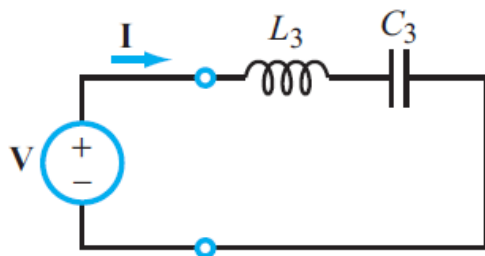
Impedance combination for RLC Circuit



(a) RL



(b) RC



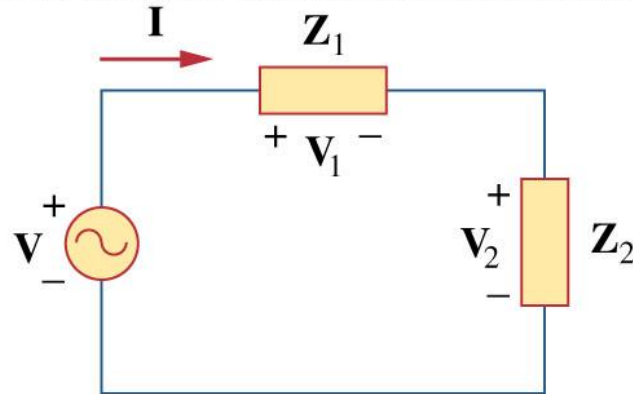
(c) LC



Voltage Divider

- Two elements in series can act like a voltage divider

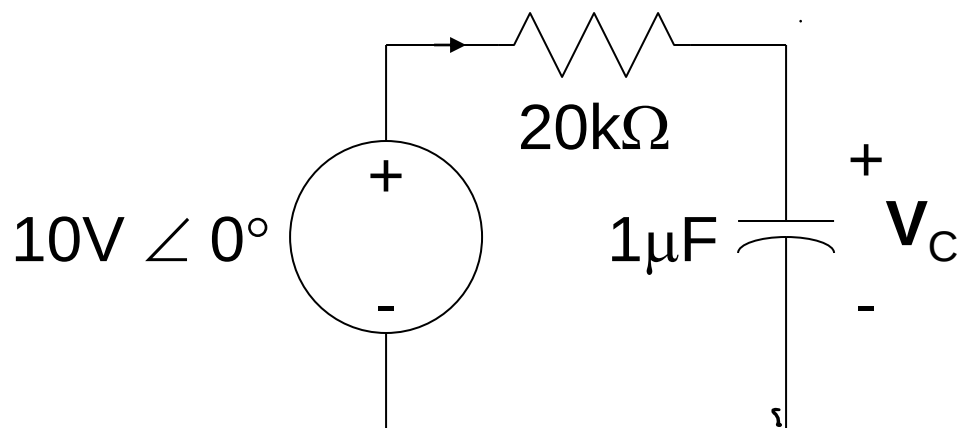
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$$\dot{V}_1 = \frac{Z_1}{Z_1 + Z_2} \dot{V} \quad \dot{V}_2 = \frac{Z_2}{Z_1 + Z_2} \dot{V}$$



Example

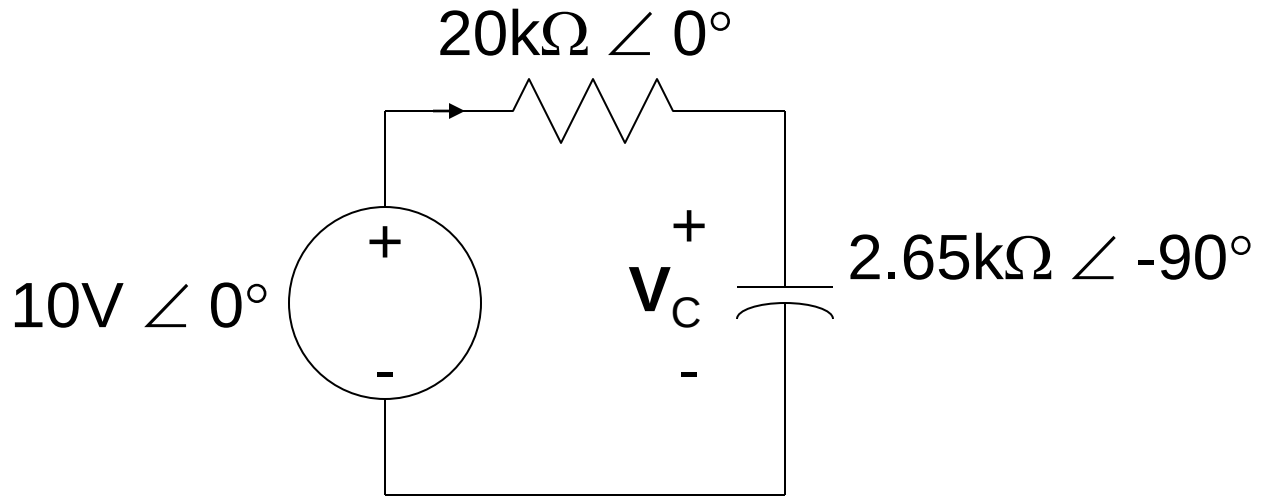


1. $f=60\text{ Hz}$, $V_C=?$

First compute impedances for resistor and capacitor:

$$Z_R = R = 20k\Omega = 20k\Omega \angle 0^\circ$$

$$Z_C = 1/j(2\pi f \times 1\mu F) = 2.65k\Omega \angle -90^\circ$$



Now use the voltage divider to find V_C :

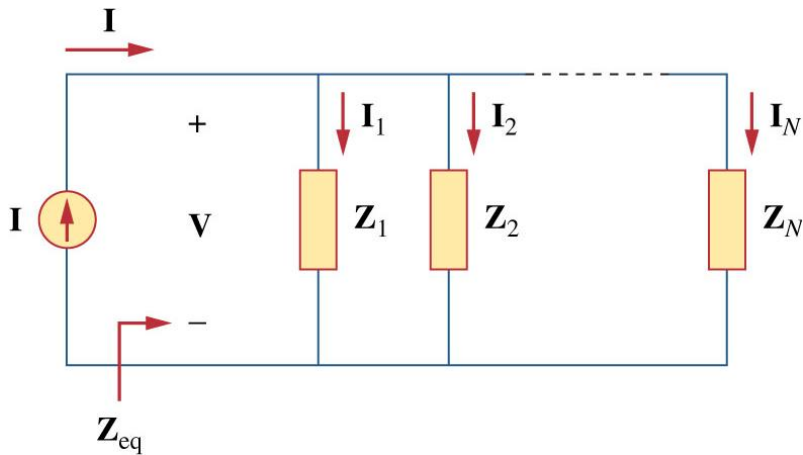
$$V_C = 10V \angle 0^\circ \left(\frac{2.65k\Omega \angle -90^\circ}{2.65k\Omega \angle -90^\circ + 20k\Omega \angle 0^\circ} \right)$$

$$V_C = 1.31V \angle -82.4^\circ$$

What if $\omega = 10$, find V_C

Parallel Combination

- Likewise, elements combined in parallel will combine in the same fashion as resistors in parallel:



$$\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} + \dots + \frac{1}{Z_N}$$

$$Y_{eq} = Y_1 + Y_2 + Y_3 + \dots + Y_N$$

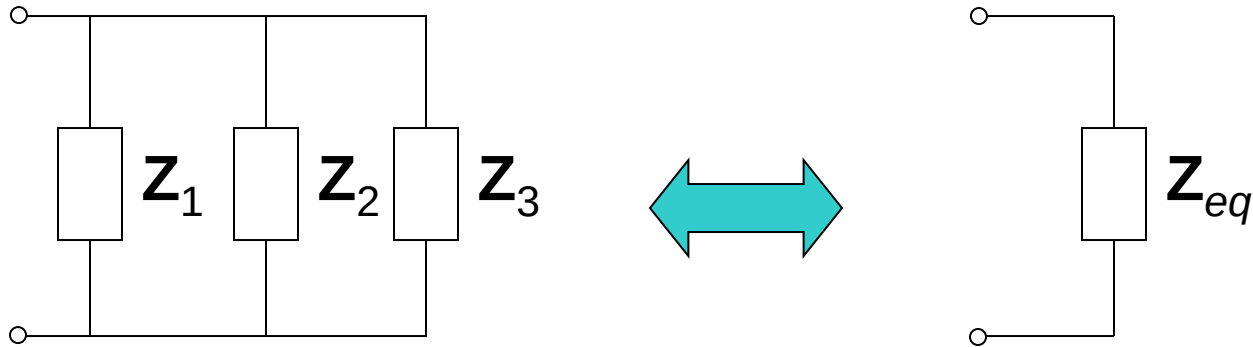
$$I_1 = \frac{Y_1}{Y_1 + \dots + Y_N} I$$

$$I_2 = \frac{Y_2}{Y_1 + \dots + Y_N} I$$

.....

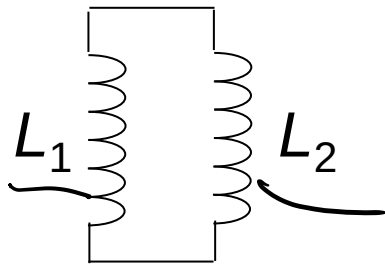


Parallel Impedance

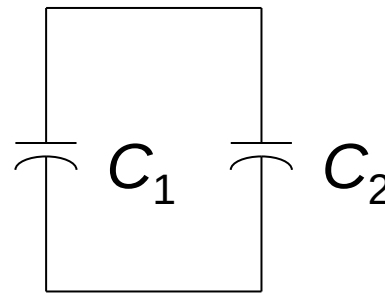


For example:

$$1/Z_{eq} = 1/Z_1 + 1/Z_2 + 1/Z_3$$



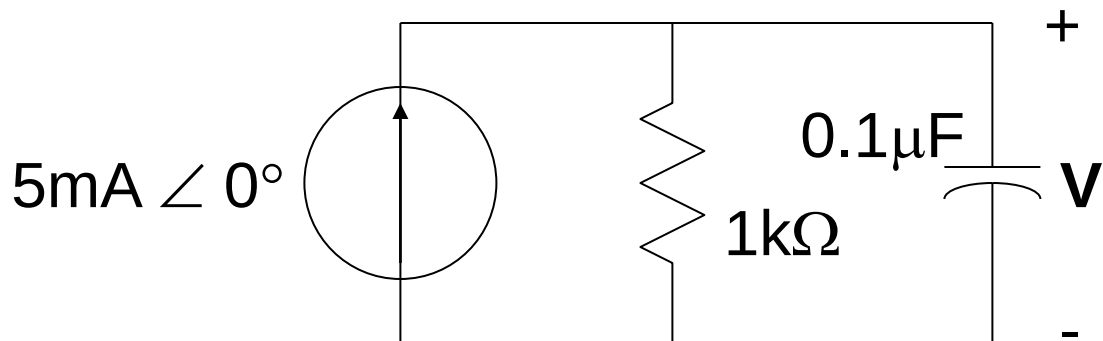
$$Z_{eq} = j\omega \frac{L_1 L_2}{(L_1 + L_2)}$$



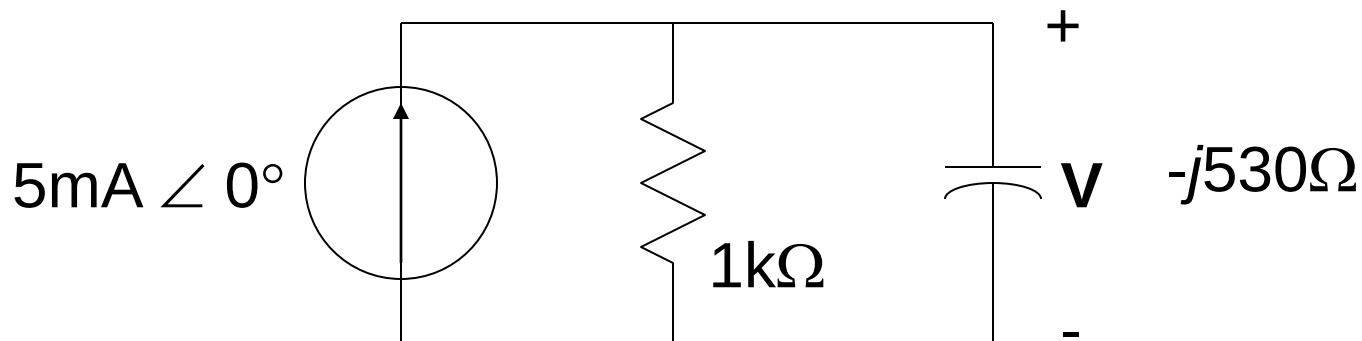
$$Z_{eq} = \frac{1}{j\omega(C_1 + C_2)}$$

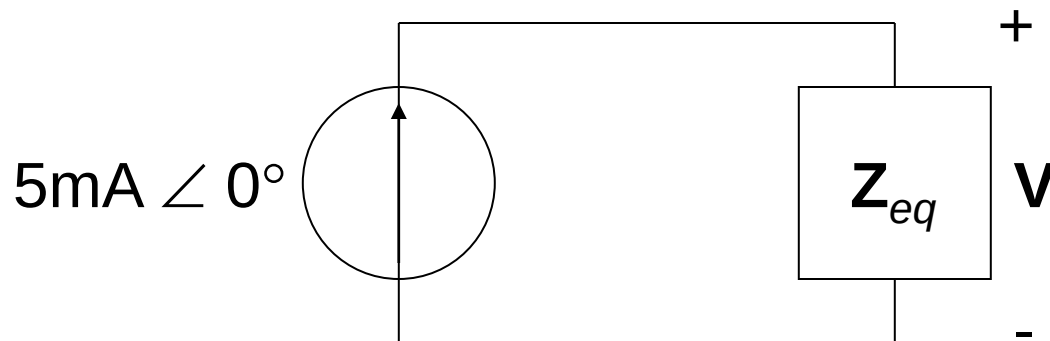


Example



Find $v(t)$ for $\omega = 2\pi \times 3000$





$$\mathbf{Z}_{eq} = \frac{1000 \times (-j530)}{1000 - j530} = \frac{10^3 \angle 0^\circ \times 530 \angle -90^\circ}{1132 \angle -27.9^\circ}$$

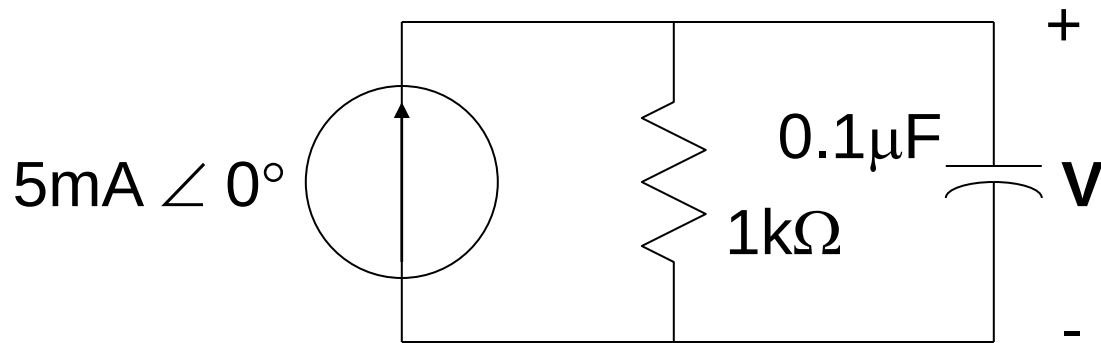
$$\mathbf{Z}_{eq} = 468.2\Omega \angle -62.1^\circ$$

$$\mathbf{V} = \mathbf{I}\mathbf{Z}_{eq} = 5\text{mA} \angle 0^\circ \times 468.2\Omega \angle -62.1^\circ$$

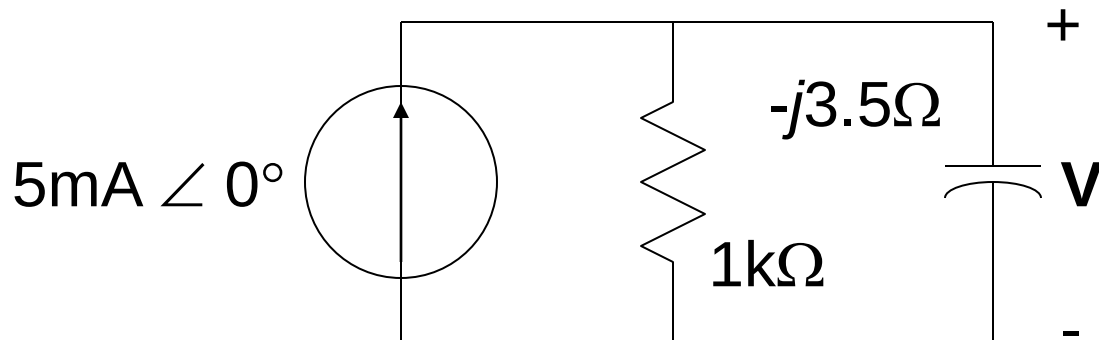
$$\mathbf{V} = 2.34\text{V} \angle -62.1^\circ$$

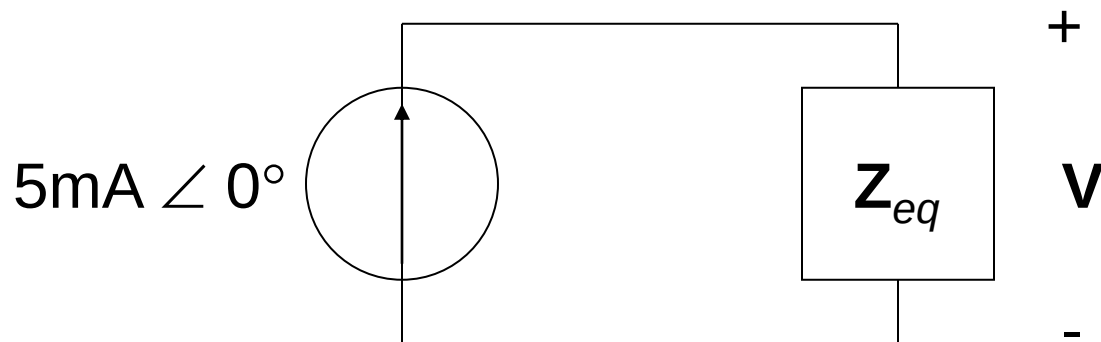
$$v(t) = 2.34 \cos(2\pi 3000t - 62.1^\circ) \text{V}$$

Change the Frequency



Find $v(t)$ for $\omega = 2\pi \cdot 455000$





$$\mathbf{Z}_{eq} = \frac{1000 \times (-j3.5)}{1000 - j3.5} = \frac{10^3 \angle 0^\circ \times 3.5 \angle -90^\circ}{1000 \angle -0.2^\circ}$$

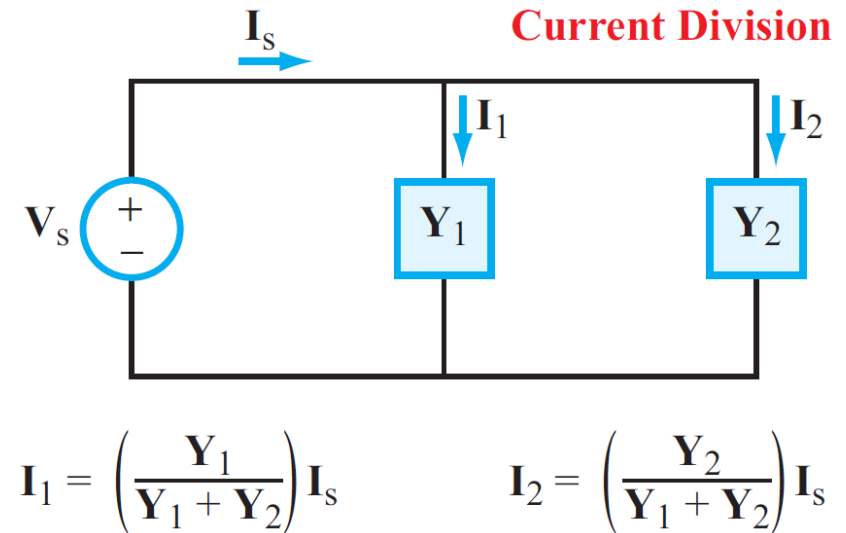
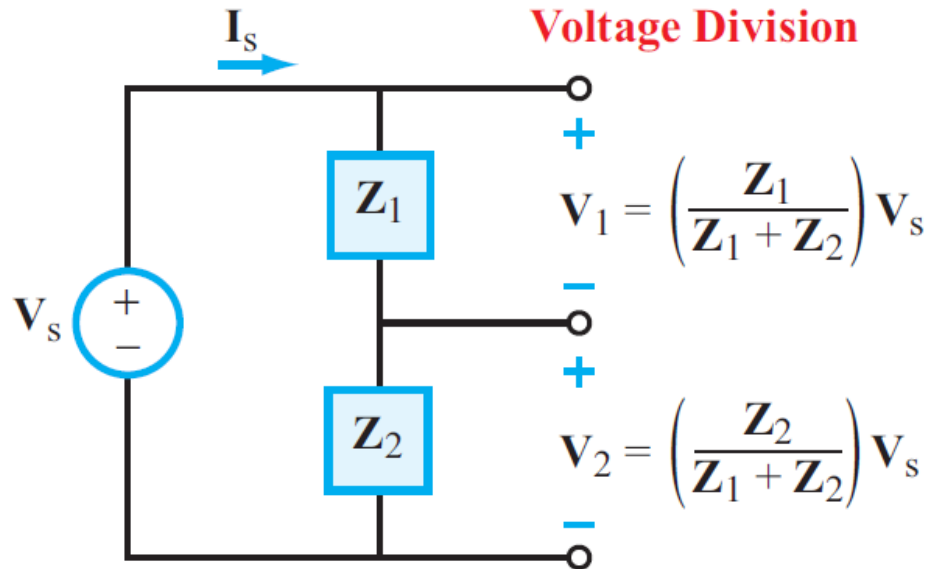
$$\mathbf{Z}_{eq} = 3.5 \angle -89.8^\circ \Omega$$

$$\mathbf{V} = \mathbf{I} \mathbf{Z}_{eq} = 5 \angle 0^\circ \text{mA} \times 3.5 \angle -89.8^\circ \Omega \quad \mathbf{V} = 17.5 \angle -89.8^\circ \text{mV}$$

$$v(t) = 17.5 \cos(2\pi 455000t - 89.8^\circ) \text{mV}$$



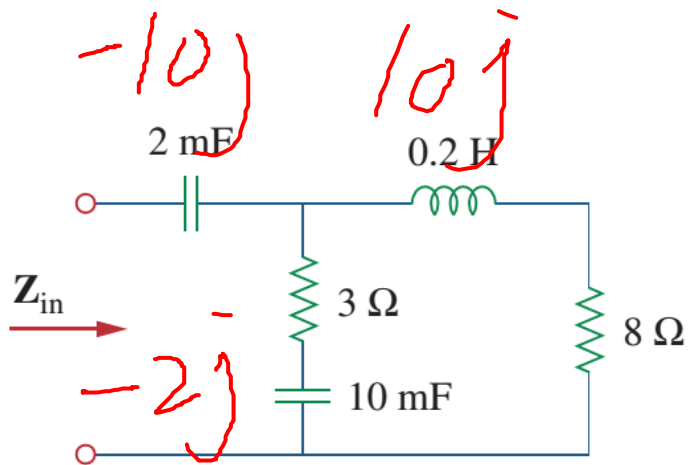
Summary: Voltage & Current Division





Exercise

- Find the input impedance of the circuit below. $\omega = 50 \text{ rad/s}$.





Outline

- Kirchhoff's laws in phasor domain
- Generalized impedance
- General AC phasor analysis
 - Nodal/mesh analysis
 - Superposition
 - Source transformation/Thevenin/Norton
- Phasor diagram



AC Phasor Analysis General Procedure

Step 1: Adopt cosine reference

$$\begin{aligned} v_s(t) &= 12 \sin(\omega t - 45^\circ) \\ &= 12 \cos(\omega t - 45^\circ - 90^\circ) = 12 \cos(\omega t - 135^\circ) \text{ V.} \\ \mathbf{V}_s &= 12e^{-j135^\circ} \text{ V.} \end{aligned}$$

Step 2: Transform circuit to phasor domain

Step 3: Cast KCL and/or KVL equations in phasor domain

$$\mathbf{Z}_R \mathbf{I} + \mathbf{Z}_C \mathbf{I} = \mathbf{V}_s,$$

which is equivalent to

$$\left(R + \frac{1}{j\omega C} \right) \mathbf{I} = 12e^{-j135^\circ}.$$

Step 1

Adopt Cosine Reference
(Time Domain)



Step 2

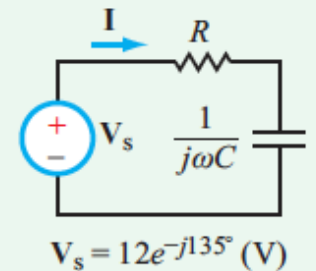
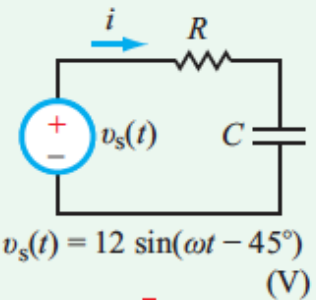
Transfer to Phasor Domain

$$\begin{aligned} i &\rightarrow \mathbf{I} \\ v &\rightarrow \mathbf{V} \\ R &\rightarrow \mathbf{Z}_R = R \\ L &\rightarrow \mathbf{Z}_L = j\omega L \\ C &\rightarrow \mathbf{Z}_C = 1/j\omega C \end{aligned}$$



Step 3

Cast Equations in
Phasor Form



$$\mathbf{I} \left(R + \frac{1}{j\omega C} \right) = \mathbf{V}_s$$



AC Phasor Analysis General Procedure

Step 4: Solve for unknown variable

$$\mathbf{I} = \frac{12e^{-j135^\circ}}{R + \frac{1}{j\omega C}} = \frac{j12\omega C e^{-j135^\circ}}{1 + j\omega RC}.$$

Using the specified values, namely $R = \sqrt{3} \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$, and $\omega = 10^3 \text{ rad/s}$,

$$\mathbf{I} = \frac{j12 \times 10^3 \times 10^{-6} e^{-j135^\circ}}{1 + j10^3 \times \sqrt{3} \times 10^3 \times 10^{-6}} = \frac{j12e^{-j135^\circ}}{1 + j\sqrt{3}} \text{ mA.}$$

$$\mathbf{I} = \frac{12e^{-j135^\circ} \cdot e^{j90^\circ}}{2e^{j60^\circ}} = 6e^{j(-135^\circ+90^\circ-60^\circ)} = 6e^{-j105^\circ} \text{ mA.}$$

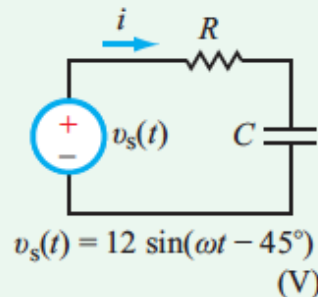
Step 5: Transform solution back to time domain

To return to the time domain, we apply the fundamental relation between a sinusoidal function and its phasor counterpart, namely

$$i(t) = \Re[\mathbf{I}e^{j\omega t}] = \Re[6e^{-j105^\circ} e^{j\omega t}] = 6 \cos(\omega t - 105^\circ) \text{ mA.}$$

Step 1

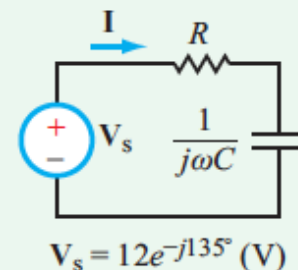
Adopt Cosine Reference
(Time Domain)



Step 2

Transfer to Phasor Domain

$$\begin{aligned} i &\rightarrow \mathbf{I} \\ v &\rightarrow \mathbf{V} \\ R &\rightarrow \mathbf{Z}_R = R \\ L &\rightarrow \mathbf{Z}_L = j\omega L \\ C &\rightarrow \mathbf{Z}_C = 1/j\omega C \end{aligned}$$



Step 3

Cast Equations in
Phasor Form

$$\mathbf{I} \left(R + \frac{1}{j\omega C} \right) = \mathbf{V}_s$$



Step 4

Solve for Unknown Variable
(Phasor Domain)

$$\mathbf{I} = \frac{\mathbf{V}_s}{R + \frac{1}{j\omega C}}$$



Step 5

Transform Solution
Back to Time Domain

$$\begin{aligned} i(t) &= \Re[\mathbf{I}e^{j\omega t}] \\ &= 6 \cos(\omega t - 105^\circ) \text{ (mA)} \end{aligned}$$

Example: RL Circuit

$$v_s(t) = 15 \sin(4 \times 10^4 t - 30^\circ) \text{ V.}$$

Also, $R = 3 \Omega$ and $L = 0.1 \text{ mH}$. Obtain an expression for the voltage across the inductor.

Solution:

Step 1: Convert $v_s(t)$ to the cosine reference

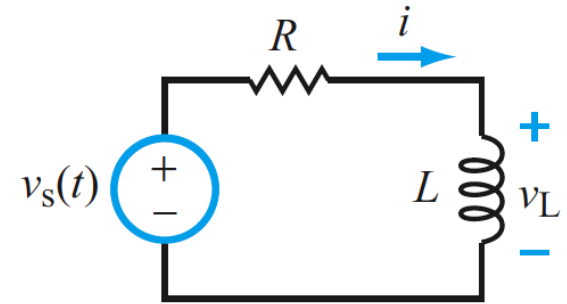
$$v_s(t) = 15 \sin(4 \times 10^4 t - 30^\circ) = 15 \cos(4 \times 10^4 t - 120^\circ) \text{ V,}$$

$$\mathbf{V}_s = 15e^{-j120^\circ} \text{ V.}$$

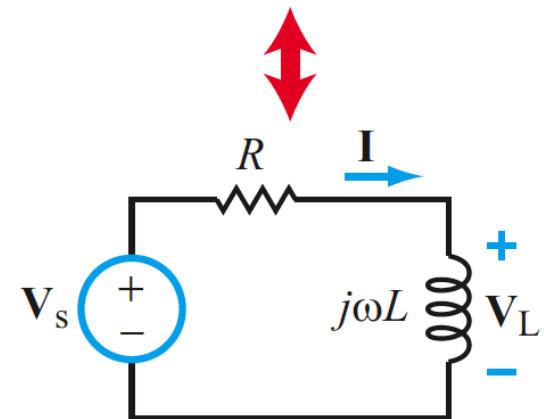
Step 2: Transform circuit to the phasor domain

Step 3: Cast KVL in phasor domain

$$R\mathbf{I} + j\omega L\mathbf{I} = \mathbf{V}_s.$$



(a) Time domain



(b) Phasor domain

Step 4: Solve for unknown variable

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}_s}{R + j\omega L} = \frac{15e^{-j120^\circ}}{3 + j4 \times 10^4 \times 10^{-4}} \\ &= \frac{15e^{-j120^\circ}}{3 + j4} = \frac{15e^{-j120^\circ}}{5e^{j53.1^\circ}} = 3e^{-j173.1^\circ} \text{ A.} \end{aligned}$$

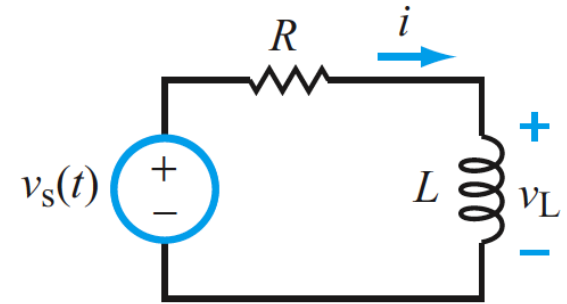
The phasor voltage across the inductor is related to $\mathbf{I}$ by

$$\begin{aligned} \mathbf{V}_L &= j\omega L \mathbf{I} \\ &= j4 \times 10^4 \times 10^{-4} \times 3e^{-j173.1^\circ} \\ &= j12e^{-j173.1^\circ} \\ &= 12e^{-j173.1^\circ} \cdot e^{j90^\circ} = 12e^{-j83.1^\circ} \text{ V,} \end{aligned}$$

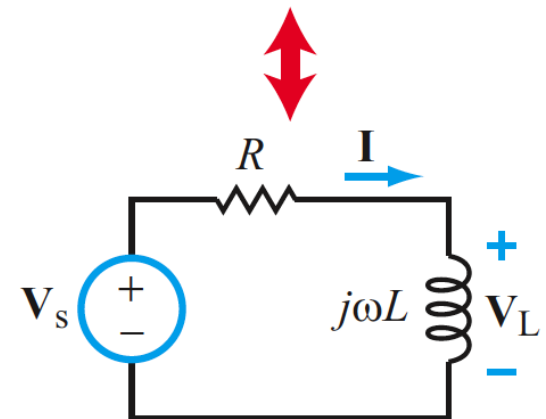
where we replaced j with e^{j90° .

Step 5: Transform solution to the time domain

$$\begin{aligned} v_L(t) &= \Re[\mathbf{V}_L e^{j\omega t}] \\ &= \Re[12e^{-j83.1^\circ} e^{j4 \times 10^4 t}] \\ &= 12 \cos(4 \times 10^4 t - 83.1^\circ) \text{ V.} \end{aligned}$$



(a) Time domain

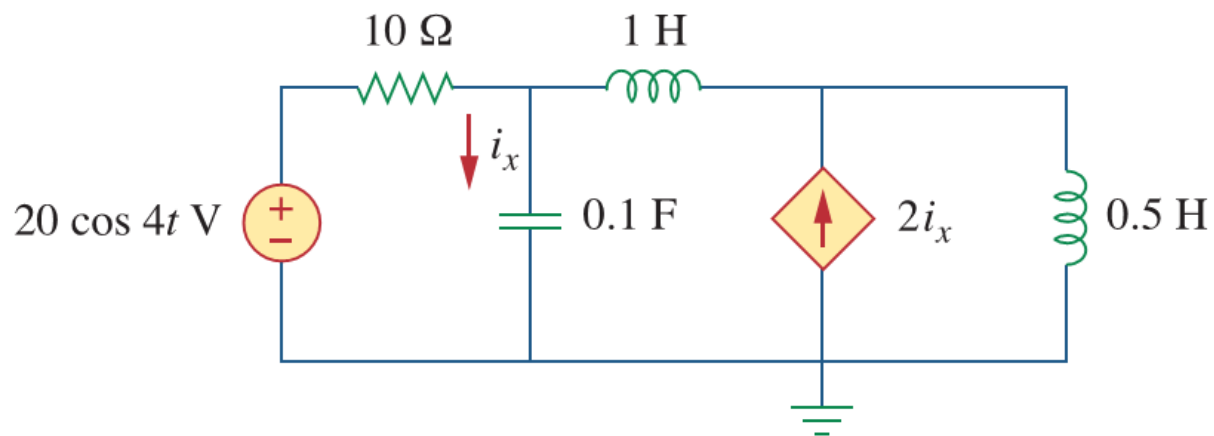


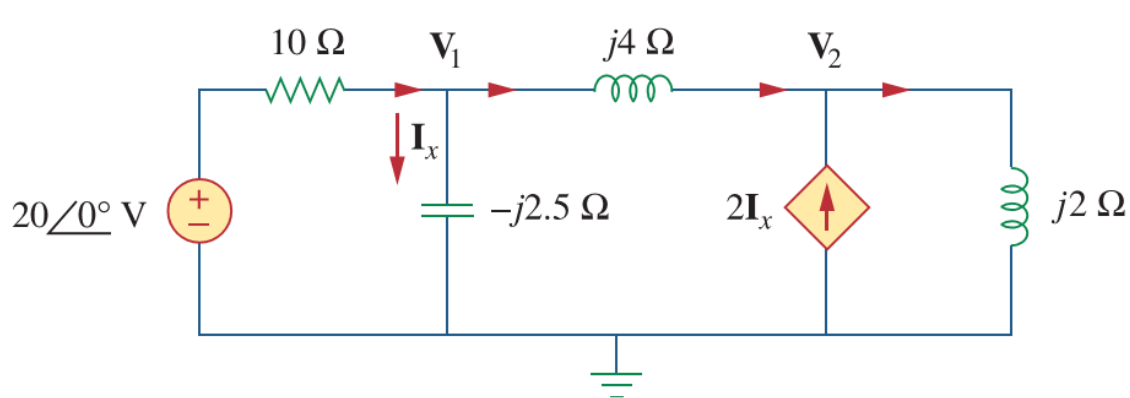
(b) Phasor domain



Nodal Analysis

- Example---Find i_x





$$\frac{20 - V_1}{10} = \frac{V_1}{-j2.5} + \frac{V_1 - V_2}{j4}$$
$$2\mathbf{I}_x + \frac{V_1 - V_2}{j4} = \frac{V_2}{j2}$$

$$i_x = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$



10.5 Find i_o in the circuit of Fig. 10.54.

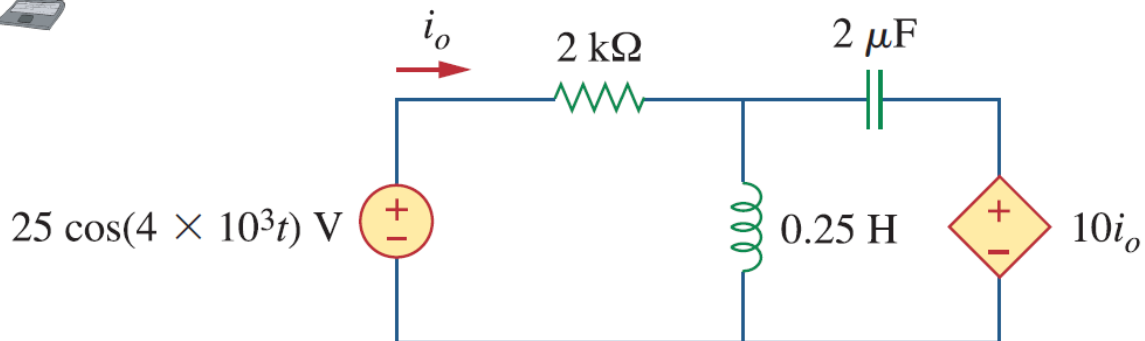
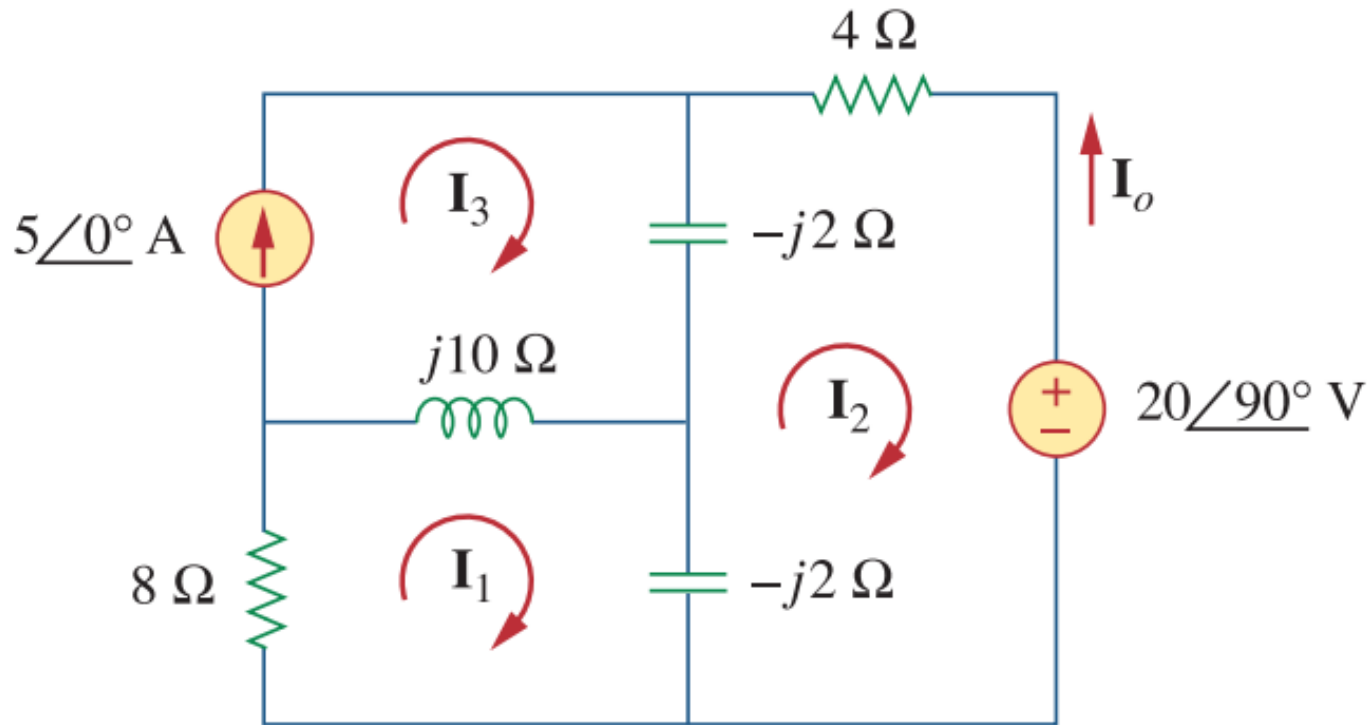


Figure 10.54

For Prob. 10.5.

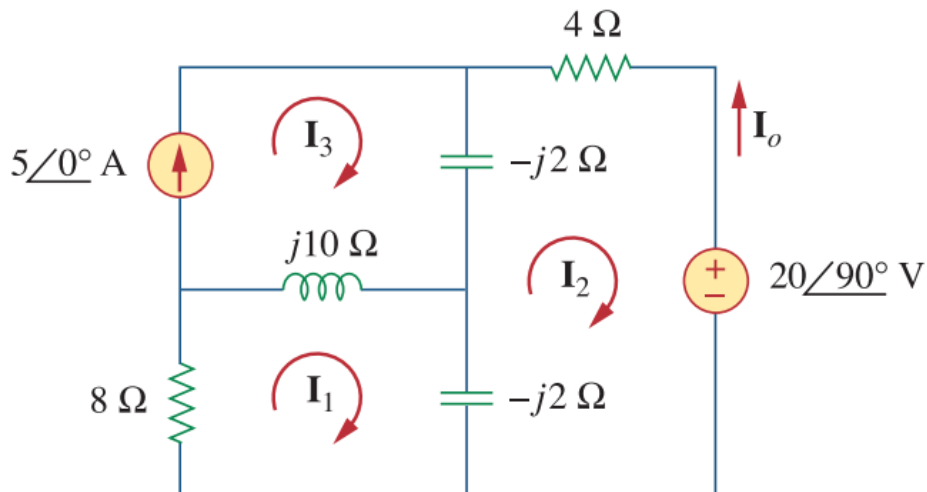


Mesh Analysis





Mesh Analysis



Applying KVL to mesh 1, we obtain

$$(8 + j10 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - j10\mathbf{I}_3 = 0 \quad (10.3.1)$$

For mesh 2,

$$(4 - j2 - j2)\mathbf{I}_2 - (-j2)\mathbf{I}_1 - (-j2)\mathbf{I}_3 + 20\angle 90^\circ = 0 \quad (10.3.2)$$

For mesh 3, $\mathbf{I}_3 = 5$. Substituting this in Eqs. (10.3.1) and (10.3.2), we get

$$(8 + j8)\mathbf{I}_1 + j2\mathbf{I}_2 = j50 \quad (10.3.3)$$

$$j2\mathbf{I}_1 + (4 - j4)\mathbf{I}_2 = -j20 - j10 \quad (10.3.4)$$



10.27 Using mesh analysis, find $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit of Fig. 10.75.

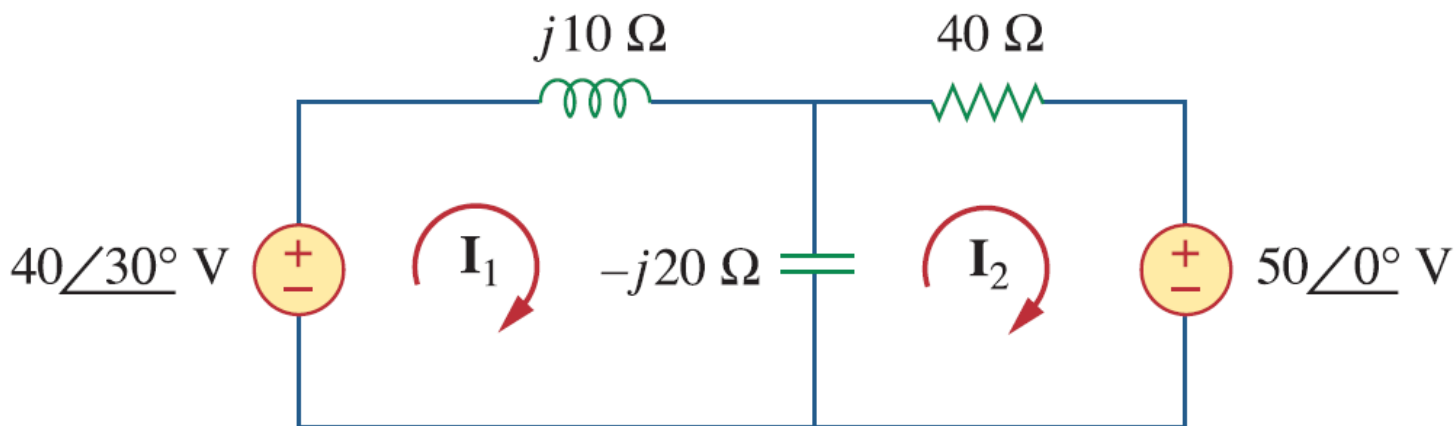
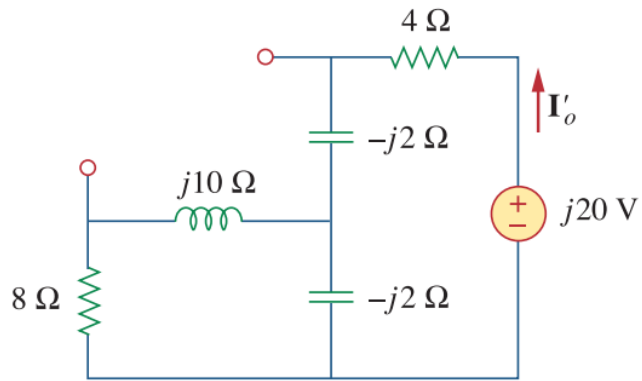
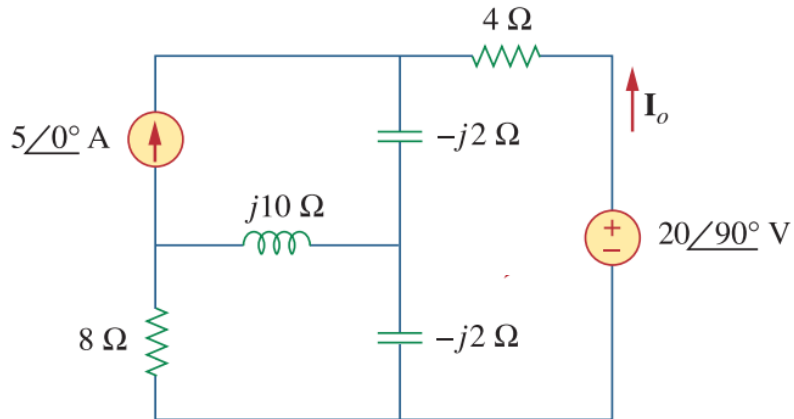


Figure 10.75

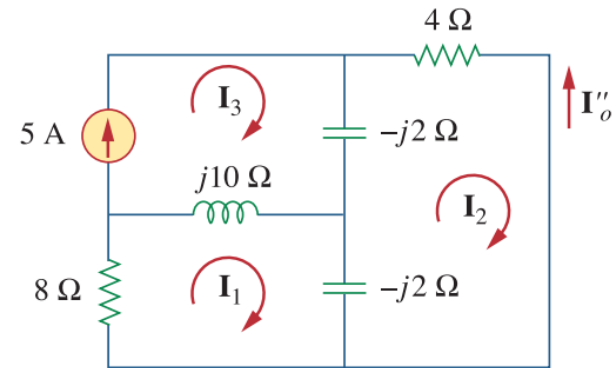
For Prob. 10.27.



Superposition-Example



(a)



(b)

Multiple Frequencies

10.46 Solve for $v_o(t)$ in the circuit of Fig. 10.91 using the superposition principle.

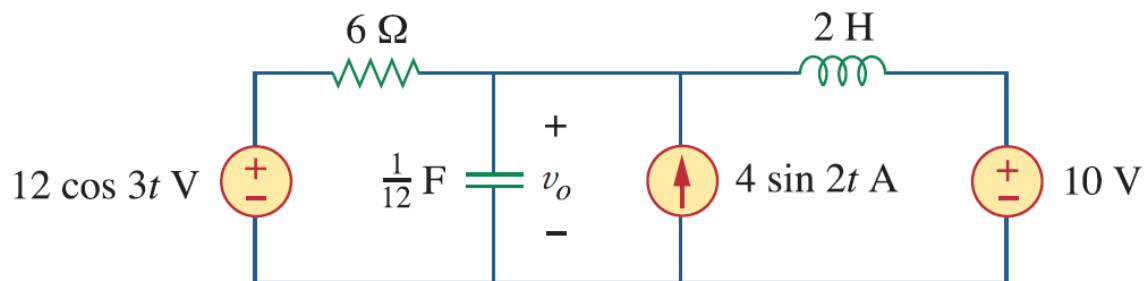


Figure 10.91
For Prob. 10.46.



10.48 Find i_o in the circuit of Fig. 10.93 using superposition.



ML

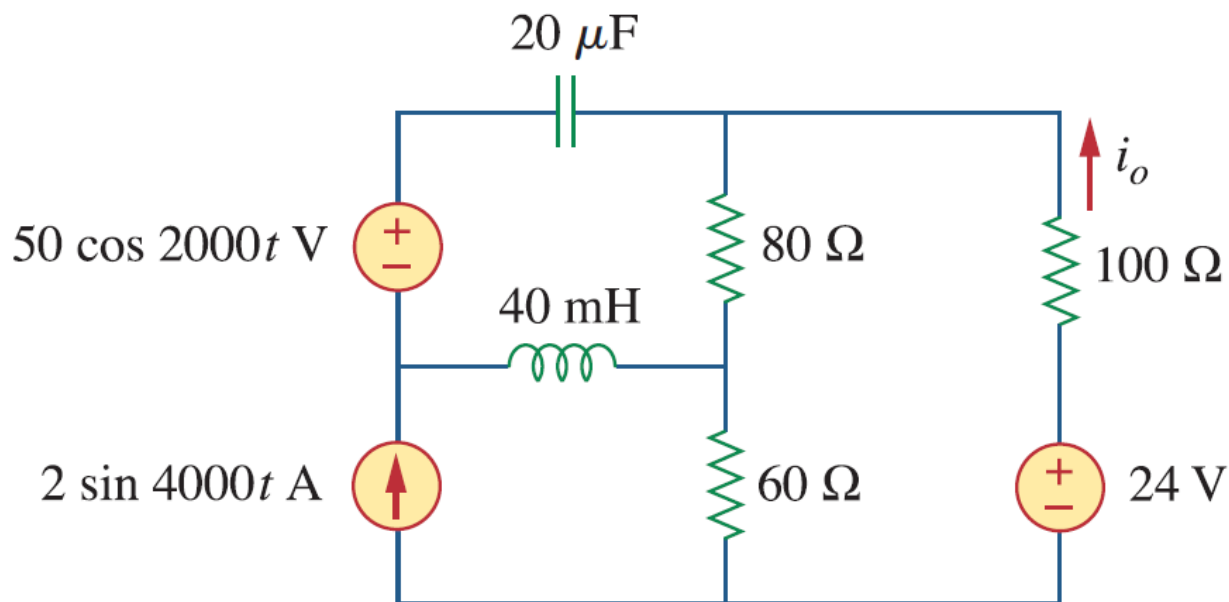
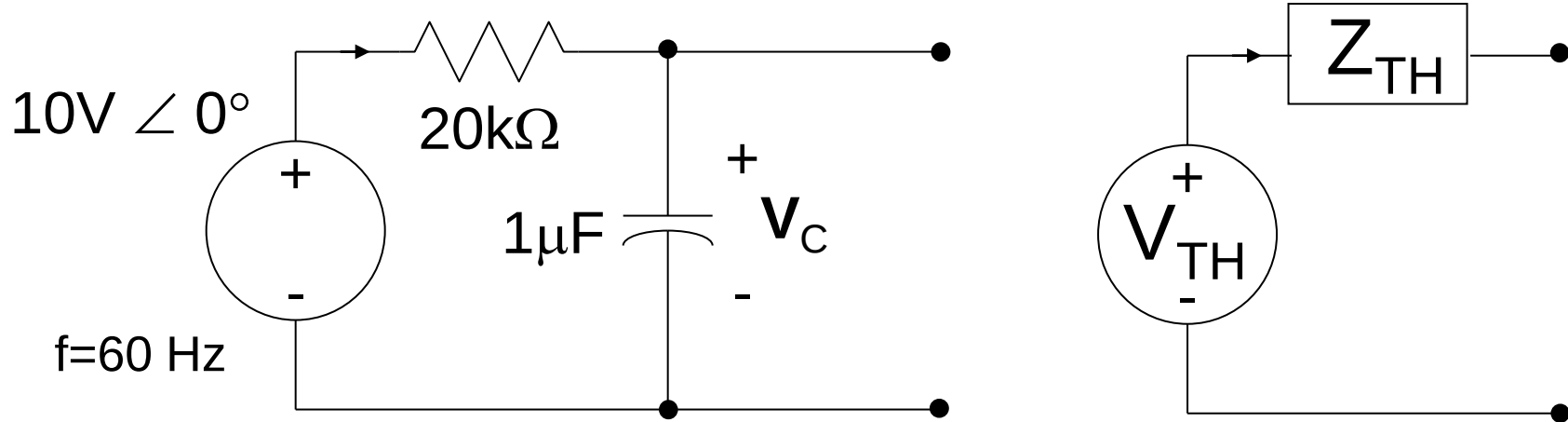


Figure 10.93

For Prob. 10.48.



Thevenin Equivalent



$$\mathbf{Z}_R = R = 20\text{k}\Omega = 20\text{k}\Omega \angle 0^\circ$$

$$\mathbf{Z}_C = 1/j(2\pi f \times 1\mu\text{F}) = 2.65\text{k}\Omega \angle -90^\circ$$

$$\mathbf{V}_{TH} = \mathbf{V}_{OC} = 10\text{V} \angle 0^\circ \left(\frac{2.65\text{k}\Omega \angle -90^\circ}{2.65\text{k}\Omega \angle -90^\circ + 20\text{k}\Omega \angle 0^\circ} \right) = 1.31 \angle -82.4$$

$$\mathbf{Z}_{TH} = \mathbf{Z}_R \parallel \mathbf{Z}_C = \left(\frac{20\text{k}\Omega \angle 0^\circ \cdot 2.65\text{k}\Omega \angle -90^\circ}{2.65\text{k}\Omega \angle -90^\circ + 20\text{k}\Omega \angle 0^\circ} \right) = 2.62 \angle -82.4$$



10.67 Find the Thevenin and Norton equivalent circuits at terminals a - b in the circuit of Fig. 10.110.



ML

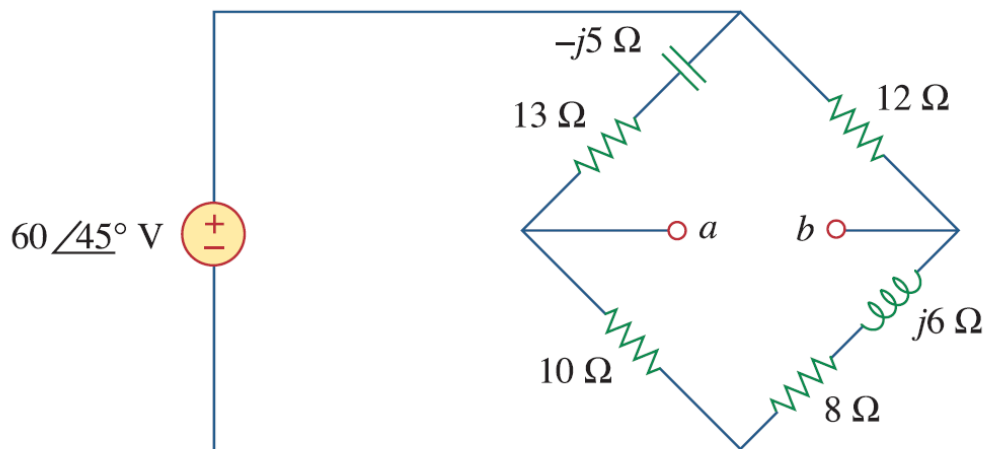
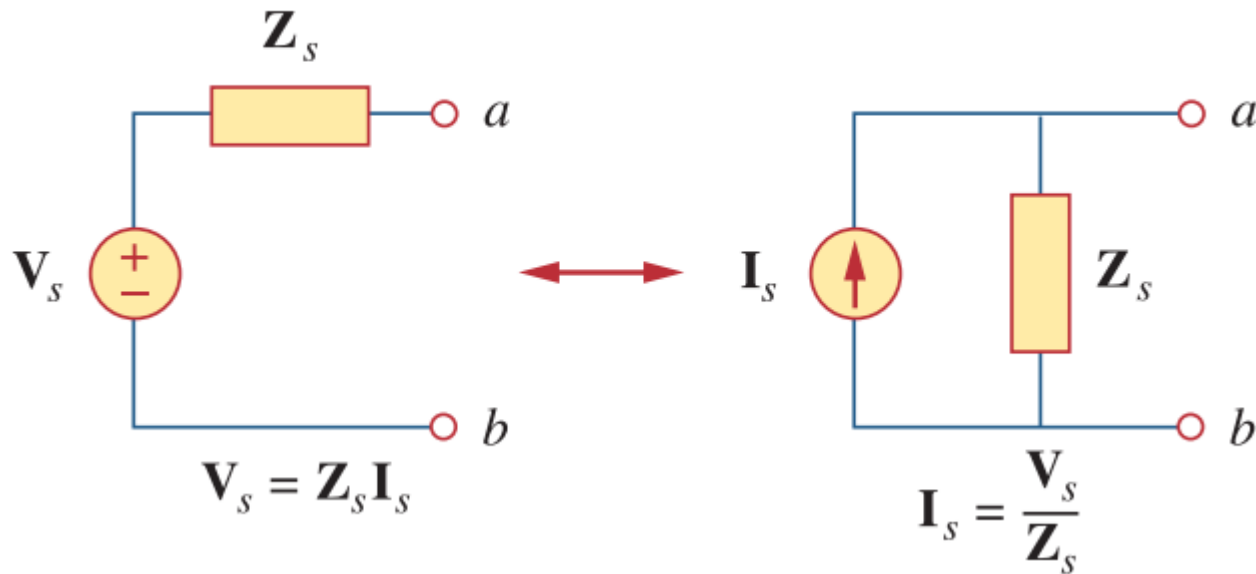


Figure 10.110

For Prob. 10.67.



Source transformation/Norton



$$V_s = Z_s I_s \quad \Leftrightarrow \quad I_s = \frac{V_s}{Z_s}$$

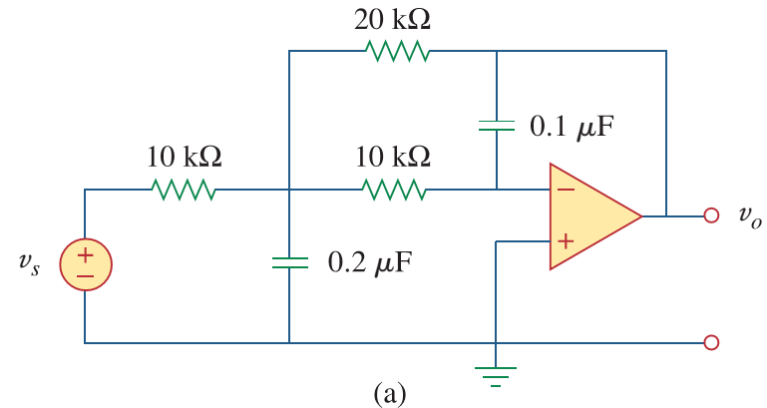
AC Op Amp Circuits

Question 1: Are op amps used in ac circuits?

Answer 1: Yes.

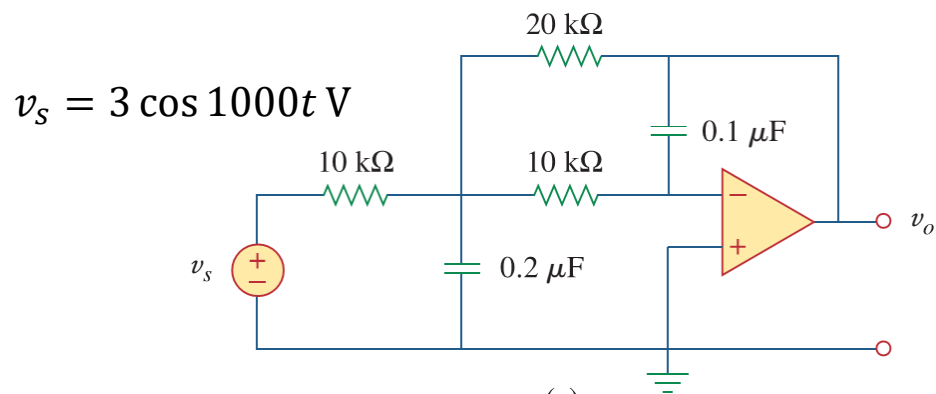
Question 2: Is the ideal op-amp model applicable to ac circuits?

Answer 2: The ideal op-amp model is based on the assumption that the open-loop gain A is very large ($> 10^4$), which is true at dc and low frequencies, but not necessarily so at high frequencies. The range of frequencies over which A is large depends on the specific op-amp design.

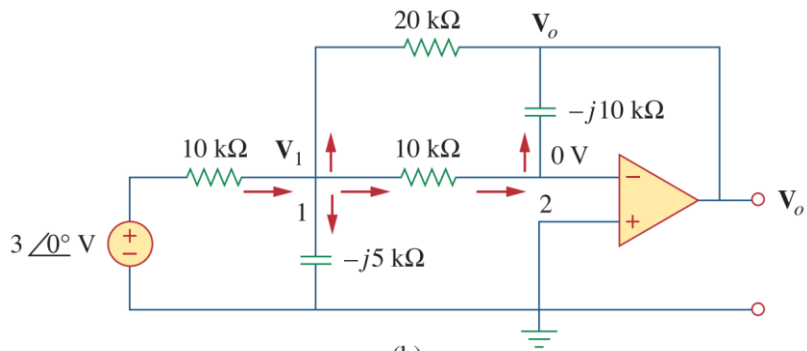
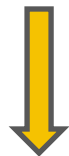




Example –find v_o



(a)



(b)

10.74 Evaluate the voltage gain $\mathbf{A}_v = \mathbf{V}_o/\mathbf{V}_s$ in the op amp circuit of Fig. 10.117. Find $\mathbf{A}_v$ at $\omega = 0$, $\omega \rightarrow \infty$, $\omega = 1/R_1C_1$, and $\omega = 1/R_2C_2$.

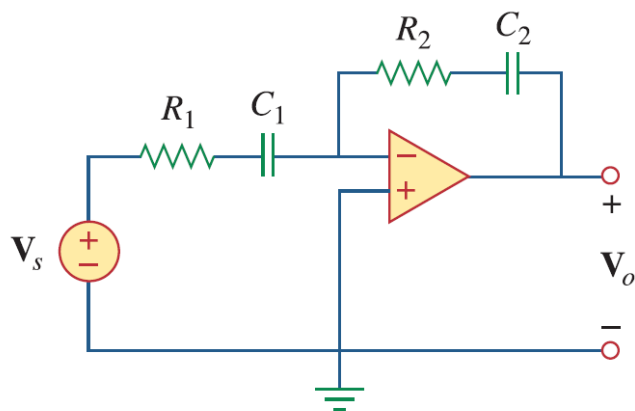


Figure 10.117

For Prob. 10.74.



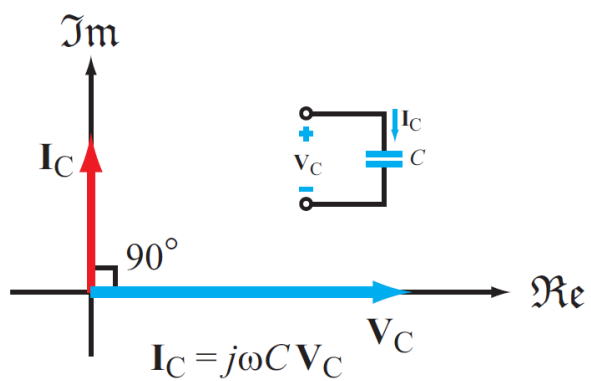
Outline

- Kirchhoff's laws in phasor domain
- Generalized impedance
- General AC phasor analysis
- Phasor diagram

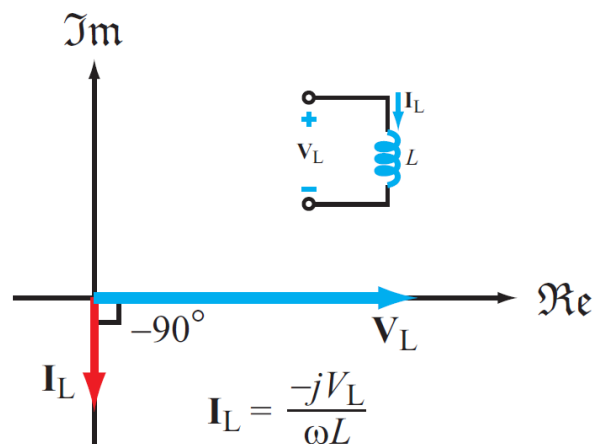


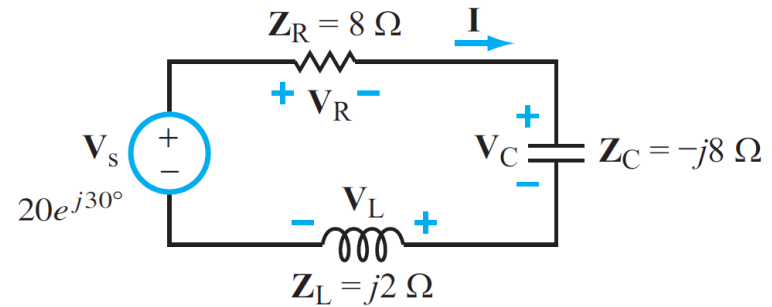
Phasor Diagrams

Capacitor

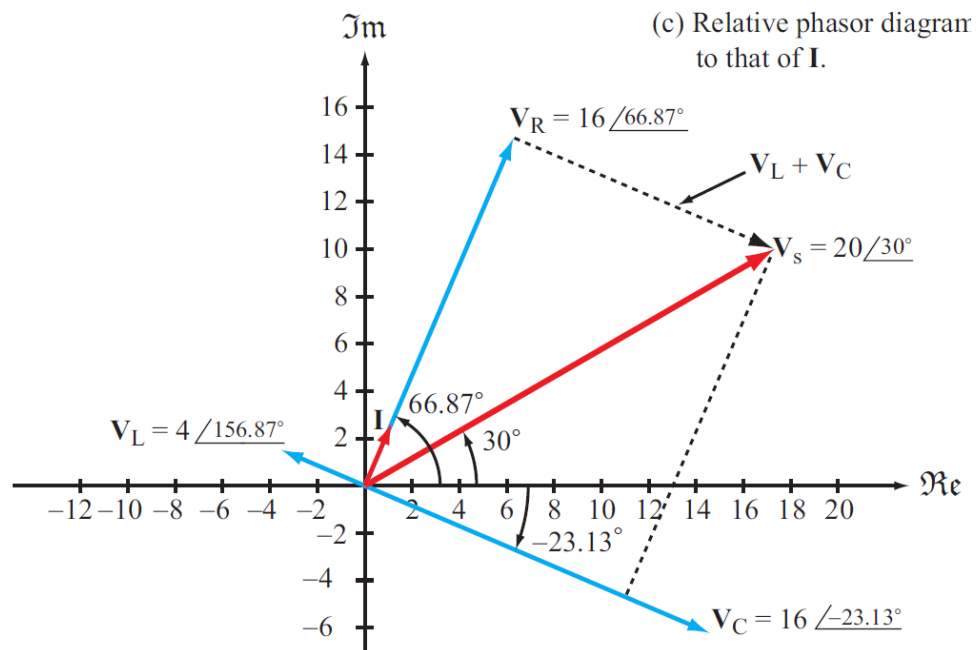


Inductor





$$\mathbf{I} = \frac{\mathbf{V}_s}{R + j\omega L - \frac{j}{\omega C}} = \frac{20e^{j30^\circ}}{8 + j2 - j8} = \frac{20e^{j30^\circ}}{8 - j6} = \frac{20e^{j30^\circ}}{10e^{-j36.87^\circ}} = 2e^{j66.87^\circ} \text{ A}$$



10.79 For the op amp circuit in Fig. 10.122, obtain $v_o(t)$.

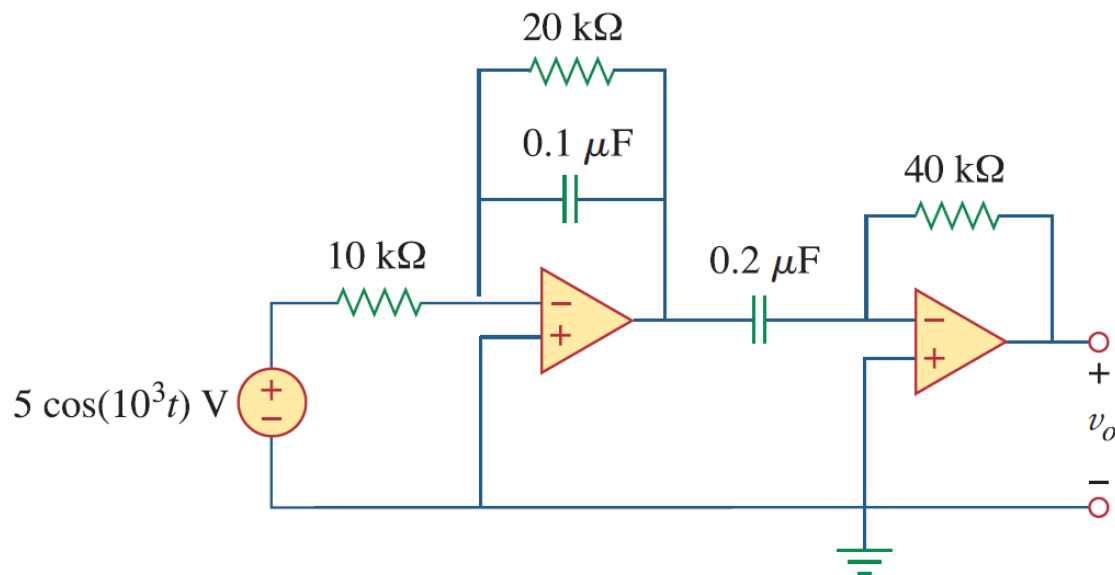


Figure 10.122

For Prob. 10.79.

10.89 The op amp circuit in Fig. 10.131 is called an *inductance simulator*. Show that the input impedance is given by

$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{V}_{\text{in}}}{\mathbf{I}_{\text{in}}} = j\omega L_{\text{eq}}$$

where

$$L_{\text{eq}} = \frac{R_1 R_3 R_4}{R_2} C$$

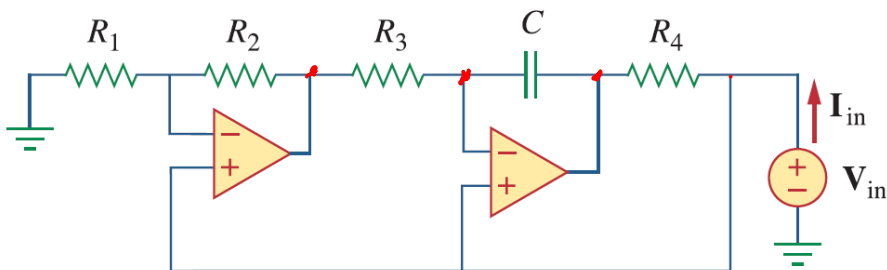


Figure 10.131

For Prob. 10.89.



10.91 Consider the oscillator in Fig. 10.133.

- (a) Determine the oscillation frequency.
- (b) Obtain the minimum value of R for which oscillation takes place.

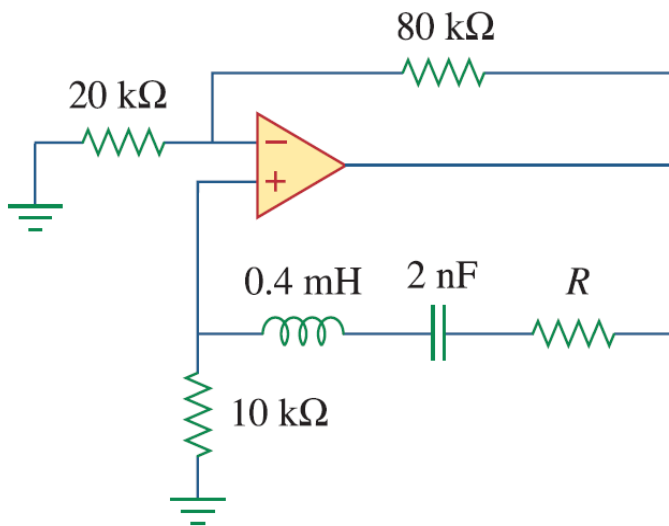


Figure 10.133

For Prob. 10.91.